# Aayush Kumar Singh

Full Stack Developer
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Portfolio | GitHub | LinkedIn

## Professional Summary

Full Stack Developer with hands-on experience building production web applications, AI-powered learning platforms, and payment systems using React, Next.js, Node.js, Express.js, and FastAPI. Experienced in end-to-end payment integration, REST API development, database design, and RAG-based AI applications. Built platforms that reduced reported tutor search time by 75% and manual study preparation time by 70%. Hackathon runner-up with Agile team collaboration experience.

## Education

**Institute of Engineering, Purwanchal Campus** 2022 – 2026
*Bachelor in Computer Engineering*

## Experience

**Full Stack Developer Intern** Jul. 2026 – Present
*NavaNexa Tech Pvt. Ltd.*

- Own end-to-end payment engineering across company products, from R&D, API evaluation, and workflow design to gateway integration, implementation, and testing with CityPay and Fonepay.
- Develop and integrate 50+ API endpoints across nchat payment workflows, covering KYC, order payments, peer-to-peer (P2P) transfers, and peer-to-merchant (P2M) transfers.
- Designed and built the bookkeeping system from initial research and requirements analysis through implementation, supporting centralized payment workflows and transaction management across products.
- Integrated AI Builder subscription payments with the bookkeeping system and implemented configurable controls for KYC and order-payment functionality.

**Web & Mobile Application Development Intern** Jan. 2026 – Apr. 2026
*Nexalaris Tech Pvt. Ltd.*

- Led a 4-person team (Team Sphinx) through Agile/Scrum sprints, coordinating daily standups and stakeholder discussions to deliver AI-integrated software.
- Developed the *Himalayan Bodhi Spices* e-commerce platform using Next.js, TypeScript, Tailwind CSS, and Framer Motion, delivering 10+ responsive pages with consistent layouts and reusable UI patterns.
- Designed and documented 10+ REST API endpoints for authentication and user management using Postman; tested the Subhyatra React Native application across 3 release cycles, identifying 20+ bugs through functional, UI, and regression testing.

## Projects

**TuitionMaster – Tutoring Marketplace** | *React, Node.js, Express.js, MongoDB, Gemini AI, Cloudinary, Docker* Live Site

- Built a full-stack tutoring marketplace featuring JWT and Google OAuth authentication, a multi-step tutor profile wizard, Cloudinary uploads, moderated reviews, and an admin panel for tutor verification and analytics.
- Reduced reported tutor search time by **75%** with a Gemini-powered assistant supporting natural-language tutor discovery, live profile retrieval, and tutor-request posting.

**SmartPrep AI – Virtual Teaching Assistant** | *Next.js, FastAPI, MongoDB, ChromaDB, RAG, LLM, NLP* Live Demo

- Developed a full-stack learning platform with PDF uploads, multi-document ingestion, user sessions, automated assessments, progress tracking, and OpenAPI/Swagger API documentation.
- Reduced reported manual study preparation time by **70%** by combining RAG pipelines, ChromaDB vector search, and LLMs to generate summaries, lessons, and MCQs from uploaded study materials.

**Employee Management System** | *React, Node.js, Express.js, MongoDB, TypeScript, Inngest, JWT* Live Demo

- Built HR workflows for attendance, leave management, and automated payslip generation across 50+ employee profiles, reducing estimated manual HR effort by **60%**.
- Implemented role-based JWT authentication across 2 dashboards and 15+ REST API endpoints; automated notifications and email delivery using Inngest and Nodemailer.

**Soulmate – Real-Time Dating & Matchmaking App** | *React Native, Expo, Node.js, Express.js, MongoDB, WebRTC, Socket.IO, Cloudinary* GitHub

- Engineered a full-stack cross-platform dating app featuring gesture-driven card swiping, JWT authentication, push notifications, real-time chat via Socket.IO, and peer-to-peer audio/video calling using WebRTC.
- Architected a modular multi-stage recommendation engine combining behavioral taste modeling, reciprocal compatibility scoring, collaborative filtering, and diversity re-ranking to optimize match relevance and discovery.

## Technical Skills

**Languages**: JavaScript, TypeScript, Python, SQL
**Frontend**: React.js, React Native, Expo, Next.js, Tailwind CSS, Bootstrap, Responsive UI Development
**Backend**: Node.js, Express.js, FastAPI, REST API Design, WebSockets, Socket.IO
**Databases**: MongoDB, PostgreSQL (Neon), MySQL, ChromaDB, Vector Search
**AI/ML**: RAG, LLM Integration, NLP, Semantic Search, scikit-learn, MobileNet
**Tools & Concepts**: Git, GitHub Actions, Docker, Postman, Jest, React Testing Library, Playwright, Vercel, Render, Netlify, Inngest, Stripe, JWT Authentication, CI/CD, SEO, Agile/Scrum

## Achievements

**1st Runner-up – Janakpur Hackathon 2.0** Dec. 2025

- Built a civic-tech platform with a RAG-based AI chatbot grounded in Nepal's Constitution 2072, real-time notifications, and NGO categorization for incident response.

## Certifications

Understanding Machine Learning · Supervised Learning with scikit-learn · Unsupervised Learning in Python *DataCamp*